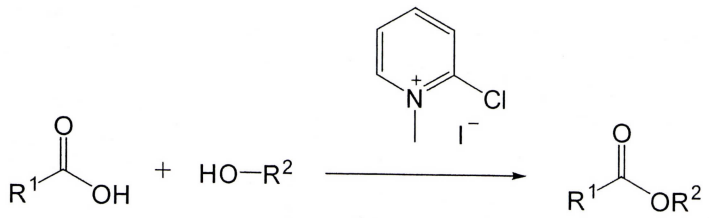
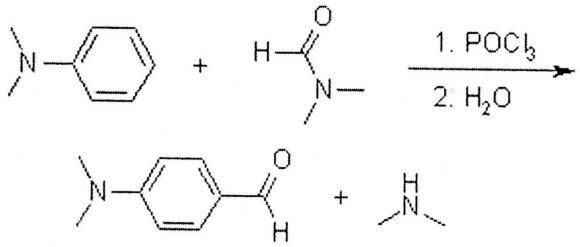
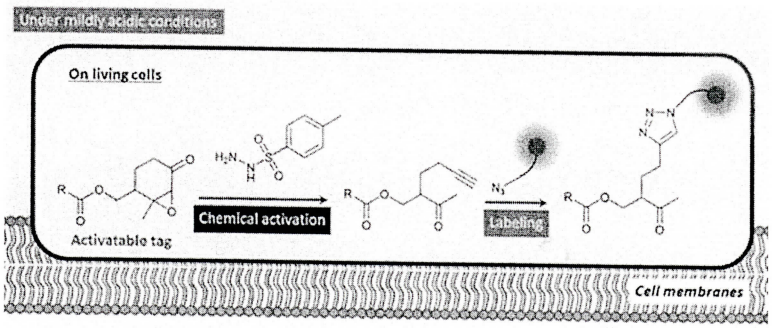
Organic Cumulative Exam
April 28, 2017

1. (20 points) Write a mechanism for the following esterification reaction.



2. (30 points) Write a mechanism for the following Vilsmeier-Haack Reaction.



3. (50 points) The first activation step in the following figure is called Eschenmoser–Tanabe Reaction. Write its mechanism. (*Bioconjugate Chemistry* **2016** *27* (9), 1976-1980)